

LD19694 Chen

Title: High-order fluctuations of temperature in hot QCD matter

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We thank Referee C for the comments and criticisms. Our response is presented as follows.

Answer to Referee C:

- **Comment:** *“This manuscript claims to introduce a theoretical framework for computing temperature fluctuations in hot QCD matter, motivated by their potential extraction from event-by-event mean transverse momentum fluctuations in heavy-ion collisions. The authors report a construction of a new thermodynamic state function $W = \Omega + TS$ and claim that its natural variables (S, V, μ_B) match the constrained quantities in experimental measurements at fixed multiplicity. By taking successive derivatives of the intensive quantity $w = W/V$ with respect to entropy density, they derive analytic expressions for temperature fluctuation cumulants c_n to arbitrary order, expressing them in terms of pressure derivatives with respect to temperature.*

The numerical results are obtained using a 2+1 flavor low-energy effective field theory within the functional renormalization group approach. The authors report that: (1) temperature fluctuations are significantly suppressed as the system transitions from the hadron resonance gas (HRG) to the quark-gluon plasma (QGP) phase, which they attribute to the substantial increase in heat capacity at higher temperatures; (2) the skewness c_3 is negative across all conditions studied; and (3) when evaluated along chemical freeze-out curves, the variance and skewness show mild energy dependence at high collision energies but increase significantly below approximately 14.5 GeV. The authors argue that these signatures are model-independent consequences of the heat capacity difference between HRG and QGP phases and propose them as experimental signatures for probing QCD thermodynamics. ”

Answer: Thanks for the clear summary of the enumerated new results obtained in this work.

- **Comment:** *“The concept of temperature fluctuations in thermodynamic systems is well established and was already introduced in Landau and Lifshitz’s statistical physics textbook, where the variance is shown to be given by T^2/C_V . The generalization to higher-order cumulants presented here is straightforward and does not, by itself, warrant publication. ”*

A: We think this comment is unfair. Indeed, we did not notice the result about the variance of temperature fluctuations in Landau and Lifshitz’s book. Thanks for pointing it out. Even though, our work still finds its own value based on the points listed as follows

- (a) The results in this work, enumerated by the referee above, are new and timely, specifically in the context of the current search for the thermodynamic contributions to the mean transverse momentum fluctuations in the heavy ion collision experiments. If the temperature fluctuations of quark-gluon plasma (QGP) could be observed finally in experiments, that would be a tremendous achievement. As the referee shows above, the formula between

the variance of temperature fluctuations and the heat capacity C_V indicates that, if the variance of temperature fluctuations could be measured, the heat capacity, or the specific heat capacity of QGP, can be extracted directly from this formula, which has never been measured reliably through other methods in heavy-ion collision experiments.

- (b) Most importantly, we predict that the skewness of thermodynamic temperature fluctuations is negative, which is in sharp contrast to the positive skewness from hydrodynamic flows. Therefore, we provide our experimental colleagues with a feasible method to distinguish the thermodynamic temperature fluctuations from non-thermodynamic $\langle p_T \rangle$ fluctuations of charged particles, such as the initial geometry fluctuations. That is the reason why in this paper a negative skewness of $\langle p_T \rangle$ fluctuations is proposed as a smoking-gun signature for the thermodynamic temperature fluctuations.
 - (c) Recently, The STAR collaboration has presented the preliminary measurements of $\langle p_T \rangle$ fluctuations for different collision energies at Quark Matter 2025. One can see that all the fluctuations of different orders, from the second to the fourth, decreases significantly from the peripheral to central collisions, which is consistent with one of our theoretical predictions: “(1) temperature fluctuations are significantly suppressed as the system transitions from the hadron resonance gas (HRG) to the quark-gluon plasma (QGP) phase, which they attribute to the substantial increase in heat capacity at higher temperatures”. In the most central collisions, where the effect of initial geometry fluctuations is assumed to be minimized, the skewness of $\langle p_T \rangle$ fluctuations measured in experiments for the moment shows a trend of a negative value, but it is still consistent with zero within errors. More detailed data analyses by our experimental colleagues are in progress, and we are looking forward to the upcoming measurement.
 - (d) The formalism developed in the work is well suited for the studies of temperature fluctuations of any orders, specifically tailor-made for the heavy-ion collisions, even being dependent on the method of data analyses in experiments (we will elaborate on this point below).
- **Comment:** “I would nonetheless recommend publication if the authors could demonstrate that the introduced temperature fluctuations can indeed be measured experimentally, as they claim when stating that they have introduced “a newly introduced thermodynamic state function that directly connects to the mean transverse momentum fluctuations measured in heavy-ion collisions.” However, this connection is not established convincingly. First, it is unclear why one must fix the chemical potential μ_B in the experimental context. More fundamentally, transverse momentum fluctuations measured in experiment receive substantial contributions from resonance decays and radial flow. None of these effects are estimated in this paper. To properly address this, one would need to perform at least semi-realistic hydrodynamic simulations with fluctuating initial conditions and trace the fate of the thermal fluctuations through the evolution and freeze-out. It is not clear to me that thermal fluctuations dominate (away from the critical region). ”

A: The reason why we choose a thermodynamic state function, denoted by W in the paper, whose natural arguments are S , V , μ_B , depends on the experimental conditions and the method of data analyses. In the analysis of event-by-event fluctuations of mean transverse momenta of charged particles, only events of the similar charged particle multiplicity N_{ch} are chosen, and it is known that N_{ch} is approximately proportional to the entropy $S \sim N_{\text{ch}}$, so the entropy S is fixed. In the data analyses, the centrality of collisions, i.e., the collision parameter, is fixed, so the volume V is fixed. Moreover, the fixed cut of the rapidity makes the volume fixed

further. As for the μ_B , this is very similar with the case of the measurement of the net-baryon or net-proton fluctuations in the heavy-ion collision experiments. In every event, we detect only a small portion of protons and anti-protons, this is restricted by the acceptance window. The net-proton numbers fluctuate significantly from one event to another, whereas, the baryon chemical potential μ_B can be regarded as a constant. For example, in the heavy-ion collisions of high energy with the collision energy, e.g., $\sqrt{s_{NN}} = 200$ GeV or even ~ 1000 GeV in the LHC energy, the baryon chemical potential μ_B is very close to zero identically for the central collisions, see Fig.3 in Ref. [64] (arXiv:1710.09425). In contrast, for the lowest collision energy in the colliding mode at STAR $\sqrt{s_{NN}} = 7.7$ GeV, the chemical potential can be increased up to about $\mu_B \sim 400$ MeV. However, we should emphasize that in the regime of low collision energy, such as in the fixed target collision at STAR with $3 \text{ GeV} \leq \sqrt{s_{NN}} \leq 7.7$ GeV, the portion of the detected protons in comparison to the whole net protons increases significantly, and the effect of the global baryon conservation begins to play a role. This effect can be corrected by employing, e.g., the subensemble acceptance method (2003.13905), with a fixed μ_B .

Furthermore, the referee comments that this paper can be published only *“if the authors could demonstrate that the introduced temperature fluctuations can indeed be measured experimentally”*. We think that this is unfair either. It is a huge project to detect the temperature fluctuations in the heavy-ion collisions, which require different efforts from experiments, theory and simulations, etc. It would be a triumph if the temperature fluctuations could be measured in the future. It is apparently impossible to solve all these problem in one paper.

Very possibly it is the case that the thermodynamic contribution is buried in the non-thermodynamic $\langle p_T \rangle$ fluctuations of charged particles, such as the initial geometry fluctuations. It is challenging to separate the thermodynamic fluctuation from non-thermodynamic ones. The goal of our paper is to study the characteristics of thermodynamic temperature fluctuations, which makes it possible to dig out the thermodynamical contributions in the mean transverse momentum fluctuations from other contributions in future experiments. To that end, it is certainly necessary to understand all these contributions comprehensively, but this is beyond the scope of this paper.

We believe that we have achieved the goal in this paper. Several distinct characteristics of thermodynamic temperature fluctuations are found: i) We demonstrate the suppression of temperature fluctuations as the matter evolves from the HRG to QGP phase with the increase of temperature or baryon chemical potential. ii) it is found that the skewness of thermodynamic temperature fluctuations is negative, in sharp contrast to the positive skewness from hydrodynamic flows. Therefore, it is proposed in this paper that a negative skewness of $\langle p_T \rangle$ fluctuations can be used as a smoking-gun signature for the thermodynamic temperature fluctuations.

Finally, we conclude by quoting the comments on the value of our paper from Referee B: *“As far as I can tell, the result is a significant step forward and represents a welcome theoretical contribution that complements the progress in experiment. This can clear the way for further research with better models.”*